

Functional Interface

An interface with only one abstract method. Used with lambda expressions.

Why @FunctionalInterface

Ensures only one abstract method at compile-time and improves readability.

Functional Interface Compile Time

Validation happens at compile-time; runtime does not enforce it.

Marker Interface

Interface with no abstract methods used for tagging classes (e.g., Serializable).

Interface vs Abstract Class

Interface provides full abstraction and supports multiple inheritance; abstract class supports partial abstraction.

Concrete vs Abstract Class

Concrete class is fully implemented and can be instantiated; abstract class is partially implemented.

Constructor in Interface

Not allowed because interfaces cannot be instantiated.

Constructor in Abstract Class

Allowed and used to initialize common data for subclasses.

Object of Abstract Class

Cannot instantiate directly; use subclass or anonymous class.

Final Keyword

Used to restrict modification: variables cannot change, methods cannot be overridden, classes cannot be extended.

Final Classes

Examples include String, Integer, and Math; cannot be inherited.

Immutable vs Mutable

Immutable objects cannot change (String); mutable objects can change (StringBuilder).

Password Handling

Use char[] instead of String for security since it can be cleared.

String Creation Ways

Two main ways: string literal and new keyword; other methods include arrays and valueOf().

String Literal Storage

Stored in String Constant Pool inside heap memory.